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RESEARCH ARTICLE



How can governments prevent the spread of fake news? A situational cyberdeviance prevention analysis

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ABSTRACT

The production and consumption of fake news vary significantly across countries, yet research has not comprehensively examined country-level supply- and demand-side factors, limiting our ability to understand the mechanisms driving this phenomenon. Using situational crime-prevention theory as the foundation, in this study we conceptualize the factors influencing fake news propagation. Specifically, we theorize the role of *local mobile content and services* as a supply-side factor that offers situational technological opportunities for producing and disseminating fake news. On the demand side, we theorize citizens' *subjective well-being and happiness* as a factor that reduces a nation's fake news propensity. Additionally, we assess how the *quality of legal guardianship* within a nation moderates the influence of supply-side factors on fake news propensity. Adopting a mixed-methods approach, we first tested our theoretical model using a unique dataset compiled from seven independent cross-country sources covering 65 countries. To complement this analysis and obtain a more holistic understanding of the phenomenon, we conducted qualitative interviews with eight senior policy and civil society experts engaged in managing fake news at the national, regional, and international levels. These interviews not only corroborated the quantitative findings but also revealed boundary conditions that refine our theorized model. Our results provide a nuanced understanding of the interplay between supply-side and demand-side factors in curbing fake news production and dissemination, and offer practical insights for institutional bodies focused on crime prevention and policymakers tasked with designing ways to address the scourge of fake news including misinformation.

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1. Introduction

Fake news poses a major challenge for governments globally, costing an estimated \$80 billion annually in sectors like public, finance, advertising, retail, and media (Cavazos, 2019; Islam et al., 2020). In addition, fake news contributes to societal polarization and undermines public trust, negatively impacting governmental processes such as policymaking and democratic governance (OECD, 2020; Transparency International, 2019). Recognizing its significance, even citizens view combating fake news as a key government priority and are willing to incur economic costs for preventive measures, resonating with national attempts to devise effective long-term fake news prevention strategies (Foster & Bunting, 2019; Jo et al., 2022; World Economic Forum, 2024). The detrimental effects of fake news are particularly pronounced during crises, as seen during the Covid-19 pandemic, where such misinformation contributed to a significant loss of life and property by undermining governmental efforts (Adam et al., 2020; BBC, 2021; Loomba et al., 2021), and disrupting economic activities (Adam et al., 2020; Sarraf et al., 2024). Thus, repeated calls have been made to better understand

and curb the spread of fake news (Jha & Nishant, 2025; Poynter, 2020; Shirish et al., 2021).

Fake news is a form of cyberdeviance whose propensity is influenced by both the perpetrators and consumers of such misinformation but the phenomenon, similar to other burgeoning cybercrime threats (Interpol, 2020), has not been adequately studied from a criminological perspective by the IS community (Muncaster, 2020).¹ Such a criminology-based perspective enriches our sociotechnical inquiry tradition, enabling us to integrate technical and social perspectives, including those of victims and offenders. The term “fake news” has been defined in various ways by scholars (Allcott & Gentzkow, 2017; Wu et al., 2022). Here, it encompasses intentionally deceptive digital misinformation, propaganda, or disinformation that typically lacks adherence to legitimate journalistic practices or editorial checks (Shirish et al., 2021). From the Covid-19 experience, we observe that despite national efforts to combat pandemic-related fake news, there was an increase in crisis-related misinformation. However, such a surge in fake news varied across countries (Brennen et al., 2020; Knuuttila et al., 2022; Poushter et al., 2022; Poynter, 2023), indicating

the relevance of national idiosyncrasies in shaping its spread (Shirish & Kotwal, 2023), thereby making societal-level research of this phenomenon particularly valuable to research, policy, and practice (Gupta et al., 2022; Jha & Nishant, 2025). We consider the term “cyber” in deviance as an analytical necessity because fake news propensity represents a digitally mediated norm violation, the prevalence of which varies across countries due to differences in how their digital and physical spaces are structured (Yar & Steinmetz, 2023). We now build on this idea from the criminology literature.

Crimes in society are often driven by the “supply of” and “demand for” specific deviant behaviours (Cook, 1986), which are motivated either socially or individually (Chan et al., 2019; Shirish et al., 2021; So & Cheung, 2023). The propagation of fake news is thus highly contextual. On the supply side, fake news production is dependent on the local and situational circumstances, thereby allowing cyberdeviants to exploit contextualized fears and uncertainties (Gupta et al., 2022; Matook et al., 2022). Whereas on the demand side, prevailing societal perceptions of safety and life satisfaction affect fake news consumption (Osgood et al., 1996; Shirish et al., 2021; Smith, 2021). Hence, a higher subjective well-being and happiness can reduce susceptibility to fake news propagation by lowering negative emotions that fuel its spread (Gupta et al., 2022; Schoenmueller et al., 2022). Especially during a crisis period, general well-being and happiness can serve as a coping mechanism that deters the dissemination of misinformation (Carlson, 2020; Carreno et al., 2021; Nikolic-Ristanovic, 2014). Thus, nations with higher citizen happiness levels may see less fake news consumption and sharing. Leveraging a situational perspective, to better understand the role of supply- and demand-side factors in explaining the variation in fake news propensity across nations, we pose our first research question:

RQ1: To what extent do contextual factors such as local mobile content and services and citizens’ subjective well-being and happiness contribute to fake news propensity?

In addition to the contextual factors discussed above, prior research has highlighted the role of effective legal guardianship in reducing criminal deviance (Felson & Clarke, 1998). Strategies and tactics that are designed to reinforce normative commitment to principles such as the rule of law by creating legal certainty are hallmarks of a high-quality legal system. These measures can dissuade individuals from engaging in deviant behaviours (Hough et al., 2017); however, such factors are yet to be empirically examined in the context of fake news prevention. Therefore, we posit that the overall quality of a country’s legal system or the rule

of law may serve as an indicator for the efficiency of its legal guardianship in counteracting fake news (Renda, 2018), deserving further examination. Hence, we propose that the influence of the supply-side factor specifically, the relationship of local mobile content and services with fake news propensity is influenced by the quality of a nation’s legal system, indicated by the extent of the rule of law in that country. This leads to our second research question:

RQ2: To what extent does the quality of the legal system, indicated by the rule of law in a country, influence the relationship of local mobile content and services with fake news propensity in that nation?

To answer our research questions, we adopt a *situational crime prevention perspective*, which emphasizes strategies to reduce the likelihood of deviant behaviours (Clarke et al., 1976; Cohen & Felson, 1979; Smith & Clarke, 2012; Wikström & Kroneberg, 2022). This perspective is grounded in three key dimensions that either promote or prevent crimes. The first dimension is the presence of deviance-related situational opportunities. In the context of fake news, this refers to the availability of contextualized content and supporting technological infrastructure, such as *local mobile content and services*. The second dimension concerns the psychosocial situational environment where deviance occurs. In this study, this environment is represented by the collective cognitive framing of a nation’s citizens. *Citizens’ subjective well-being and happiness* signify greater life satisfaction, which may reduce the need for individuals to alleviate fears and uncertainties, these being conditions often associated with deviant behaviours. The third dimension involves managing and manipulating situational conditions to make deviance more complex, risky, or less rewarding for potential perpetrators (Clarke, 1997). Clearly, a higher quality of the legal system or the rule of law can create deterrents by increasing the risks and reducing the benefits associated with creating and spreading fake news. An efficient legal framework can also constrain the influence of situational opportunities provided by local mobile content and services.

To investigate the stated research questions, we use a sequential mixed-method research design to examine the propagation of Covid-19 linked fake news through the lens of situational crime-prevention theory. Our mixed method approach features a dominant quantitative phase followed by a less dominant qualitative phase. Initially, we analyzed country-level data for cross-country comparisons across seven different secondary databases. The quantitative analysis was complemented by interviews with eight senior policy and civil society experts, which enriched our findings and helped us to identify boundary conditions for our

theoretical model. The meta-inferences so arrived at, by combining the quantitative and qualitative findings, enable us to offer a holistic theoretical model that can be used as the point of departure for future fake news-related policy studies. In addition, our study contributes to the IS literature by providing an understanding of the proximal supply- and demand-side factors that explain the spread of fake news. We further contribute to IS research on cybercrime by identifying the enablers and inhibitors of the complex sociotechnical phenomenon of fake news (Faik et al., 2020; Shirish et al., 2021). Our study also offers several practical implications. For example, the identification of the salient role played by local mobile content and services in amplifying fake news propensity and the validation of this finding through interviews with policy experts provides a cue for governments and policymakers to holistically assess the impacts of local mobile content provision. Moreover, our results highlight the need for governments to implement strategies aimed at enhancing the subjective well-being of citizens to make them resilient to the threats posed by fake news.

2. Theoretical background

2.1. Fake news propensity

The production and transmission of fake news can be conceptualized as a form of digitally mediated communal deviance that bears similarities to folk crimes—i.e., deviant behaviours that may not always be seen as deviant by those who engage in them, because “everybody does it” (Clarke, 1997; Freilich & Newman, 2017). There is therefore a need to investigate both the situational and systemic factors that influence fake news propensity. Mobile connectivity has been shown to increase fake news propensity within nations, acting as a crucial supply-side factor in fake news cyber deviance (Shirish et al., 2021).

In this study, we ground fake news dissemination within Situational Crime Prevention (SCP) theory, which explains deviant behaviour as resulting from boundedly rational decisions made by individuals when situational opportunities are abundant, and deterrents are weak (Clarke, 1997; Cornish & Clarke, 1986). This perspective allows us to capture both intentionally and situationally induced fake news sharing at the country level.

Studies addressing the prevention of fake news victimization have often emphasized technical solutions to reduce the spread of fake news. For example, fact-checking mechanisms within social media platforms can help to stop citizens from falling victim to fake news (Schuetz et al., 2021). Interestingly, fake news consumption and transmission are not significantly influenced by demographic variables such as the age of potential victims (Abraham & Mandalaparthi,

2021). This highlights the need to explore individual-level psychological factors to gain deeper insights into the fake news phenomenon. Such an approach is consistent with recent calls for multidisciplinary investigations into the production and dissemination of fake news (George et al., 2021; Gradoń et al., 2021).

A handful of IS studies have examined individual cognitive factors that contribute to fake news victimization. For example, Pennycook et al. (2020) used experiments to study individuals’ willingness to browse and share misinformation via social media. It was found that citizens do not usually devote significant efforts to analytical reasoning, which explains the widespread propagation of fake news during times of crisis, such as the coronavirus pandemic (Pennycook et al., 2020). Another study deploys a mathematical model to simulate citizens’ efforts to resist fake news. It concludes that making citizens aware of the negative consequences of fake news reduces their risk of victimization (Khuong et al., 2020). On the other hand, driven by conformity pressures within their social networks, citizens may also weigh the social costs of not sharing fake news (Lawson et al., 2023). A recent study examined the efficacy of detecting fake news using the “wisdom of the crowd” approach, in which social media users are encouraged to report fake videos. The findings indicate that users’ detection abilities and cognitive biases can hinder accurate reporting, thereby undermining such policies’ effectiveness (Wang et al., 2022). Collectively, these findings highlight the primary role played by citizens’ cognitive and social dynamics, which may drive the consumption and spread of unverified mobile content, particularly during crises.

On similar lines, another recent study shows that economic and media freedoms within a nation foster a resilient cognitive mindset among citizens by enhancing their sense of predictability and security, and this in turn reduces the likelihood they will consume fake news, even when they have widespread access to mobile connectivity (Shirish et al., 2021). However, the same study also found that greater political freedom can contribute to a vulnerable cognitive mindset, undermining reasoning processes and increasing fake news propensity (Shirish et al., 2021). Other recent studies have highlighted similar effects (Miller et al., 2022; Schoenmueller et al., 2022). Thus, a cognitive mindset devoid of predictability and security is often marked by heightened uncertainty, fear, and anger.

While prior research often applies individual-level theories such as dual-process theory (e.g., Pennycook & Rand, 2019) and social cognition theory (e.g., Apuke & Omar, 2021) to explain unintentional fake news sharing, these frameworks primarily focus on cognitive heuristics, emotional reactivity, and social learning at the micro-level. In contrast, our chosen theoretical approach (described in later sections)

emphasizes structural conditions and situational opportunities shaping fake news propensity at the national level. Thus, we extend prior literature and posit that citizens with a healthy and happy mindset may lack the motivation to consume or spread fake news. We incorporate this theoretical reasoning into a crime-theoretic situational analysis to investigate fake news propensity across multiple nations and propose ways to prevent it.

2.2. Situational analysis of deviance

The criminology literature defines deviance as behaviour that violates norms, values, or laws, which can vary significantly across cultures and nations (Robertson, 2014). Deviant behaviour is typically disapproved of by the conventional normative standards of social control (Freilich & LaFree, 2017). Engaging in deviant behaviour (including cyberdeviance) is not simply a byproduct of individual lifestyles or circumstances (Cohen & Felson, 1979). Rather, it is often influenced by prior cognitive framing (Osgood et al., 1996). Contemporary research on cyber deviance emphasizes the need for a holistic examination of the phenomenon, explicitly addressing both the facilitators and inhibitors of deviance opportunities (Naidoo, 2020). Such an approach can inform the development of targeted interventions to mitigate the spread of specific deviant behaviours (Cornish & Clarke, 1986).

Cyber deviance is generally dependent on situational opportunities, including victimization potential, social and physical environments, and access to necessary technological resources (Naidoo, 2020; Wortley, 2001). Because the quality and quantity of deviance opportunities within a given setting shape the volume of deviance engendered (Cook, 1986), situational crime prevention theories emphasize eliminating facilitating opportunities to deter deviant behaviour (Clarke et al., 1976; Cohen & Felson, 1979; Smith & Clarke, 2012; Wikström & Kroneberg, 2022). The situational deviance prevention approach focuses on designing and managing environments to increase the effort and risk associated with committing crimes while reducing the potential rewards of deviant behaviour (Clarke, 1992). Strategies such as target-hardening, enhanced surveillance, and the implementation of controls have been applied in efforts to mitigate cyberdeviance (Willison, 2000).

Fake news is increasingly recognized as a form of cyber deviance—an act that violates normative expectations in online contexts and may lead to social harm (Yar & Steinmetz, 2023). Drawing from criminological research, deviant behaviour can arise from both intentional acts and unintentional dissemination, such as impulsive sharing or socially driven forwarding of fake news during crises (Holt et al., 2022; Leukfeldt et al., 2020). Traditional theories like deterrence theory or social learning theory have provided valuable insights

into individual-level deviance, yet they often fall short in addressing the situational and structural factors that shape national-level variation in fake news behaviour (Bossler & Holt, 2012). Recent work in cybercrime scholarship emphasizes the need to reconsider deviant online behaviour through a socio-technical lens that incorporates both intentional and unintentional actions, especially during times of social crisis (Holt & Holt, 2025; Holt et al., 2022). In their editorial titled Human Factors in Cybercrimes, Dupont and Holt (2022) call for theoretical frameworks that recognize the influence of human factors that include technology, social, and institutional factors, involving aspects that can allow us to decipher the psychological states of the masses. This is aligned with our adoption of SCP to explain structural variation in fake news propensity within nations (Dupont & Holt, 2022). Per SCP (Clarke, 1997), deviance is conceptualised as a function of boundedly rational decision-making in the presence of enabling opportunities and motivational states. SCP is especially suitable for our study because it can incorporate both wilful and inadvertent participation in fake news dissemination, offering a more inclusive lens than other theories of cybercrime (Leukfeldt et al., 2020; Yar & Steinmetz, 2023). This makes it a compelling framework for cross-national analysis of cyberdeviance, where opportunity structures, subjective psychosocial framing, and preventive legal mechanisms interact to influence deviant behaviours (Clarke, 1997; Leukfeldt et al., 2020).

Another reason why this perspective is most useful for this study is that SCP is not conceptually detached from explanations of offending. Foundational SCP scholarship explicitly emphasizes that preventive techniques are derived from prior analysis of how crimes are committed in specific situations. SCP is grounded in situational and ecological theories, especially rational choice theory and routine activity theory (Clarke, 1983, 1995, 1997, 2018; Clarke & Cornish, 1985). Similar to the aforementioned theories, this perspective conceptualizes offending as a purposive, context-dependent decision-making process shaped by opportunity structures (Clarke, 1983, 1995, 1997, 2018; Clarke & Cornish, 1985). In this sense, SCP necessarily engages with situational explanations of deviance by modelling the conditions under which offending occurs before specifying how those conditions might be altered to prevent it effectively. Thus, it is a holistic framework to leverage for research inquiries, although it is predominantly used by IS scholars when discussing preventive techniques in the context of cyber deviance (Ho et al., 2022).

In essence, SCP theories are premised on three key dimensions that either promote or prevent deviant behaviour. The first dimension is the availability of deviance-related situational opportunities (Felson & Clarke, 1998). In the context of fake news propagation, this refers to the accessibility of *local mobile content and services*.

The second dimension is the psychosocial situational environment, which allows the victimization risk for potential targets of deviant acts to be assessed (Cook, 1986). When it comes to fake news propagation, this involves targeted individuals' cognitive mindsets, including citizens' *subjective well-being and happiness*. This may reduce the likelihood of individuals seeking unverified information. The third dimension focuses on managing and manipulating conditions to make committing deviant acts more difficult and risky for potential offenders (Clarke, 1997). In the context of fake news propagation, this corresponds to the *quality of the extant legal system*, with a higher-quality system increasing the perceived risk and complexity of engaging in deviant acts (Felson & Clarke, 1998; Freilich & LaFree, 2017; Freilich & Newman, 2017).

Guided by the theoretical rationale outlined above and grounding our conceptualization in the existing IS literature on fake news, in Figure 1, we set out the supply- and demand-side factors that facilitate or constrain opportunities for cyberdeviance during crises. In the following section, we develop our research hypotheses, drawing on the theoretical framework illustrated in Figure 1.

3. Theoretical framework and hypothesis development

Conceptualizing fake news as a form of cyberdeviance, we apply situational crime prevention theory to explore the influence of three proximal situational factors on fake news propensity within nations. Specifically, we theorize and examine the relationships between fake news propensity on the one hand and *local mobile content and services, citizens' subjective*

well-being and happiness, and the *quality of the legal system* on the other.

The factor of *local mobile content and services* falls within the supply-side conditions that facilitate the creation and dissemination of fake news. Conversely, *citizens' subjective well-being* is a matter of demand-side conditions, because it reduces the risk of victimization (Freilich & LaFree, 2017). We expect the *quality of the legal system*, indicated by the rule of law, will moderate the relationship between supply-side conditions and fake news propensity. In the remainder of this section, we develop the arguments for the hypothesized relationships outlined in our theoretically driven research model, which we map out in Figure 2.

3.1. Linking local mobile content and services to fake news propensity

Mobile connectivity plays a crucial role during crises, serving as a tool that enables citizens to engage in sense-making (Mirbabaie et al., 2020; Shirish et al., 2021; van Ingen et al., 2017). However, it also attracts cyberdeviant actors who exploit this medium to disseminate misinformation (Ågerfalk et al., 2020; Chaudhuri et al., 2024; Doyle & Conboy, 2020; Laato et al., 2020). Prior research has demonstrated that mobile connectivity enables deviant actors to create and propagate fake news (Shirish et al., 2021). Using situational deviance prevention analysis, we argue that within the broader domain of mobile connectivity, the availability of contextualized local mobile content and services facilitates the creation and dissemination of fake news.

Fake news is a form of cyberdeviance that does not require actors to breach cybersecurity systems. Instead, it relies on social engineering techniques designed to

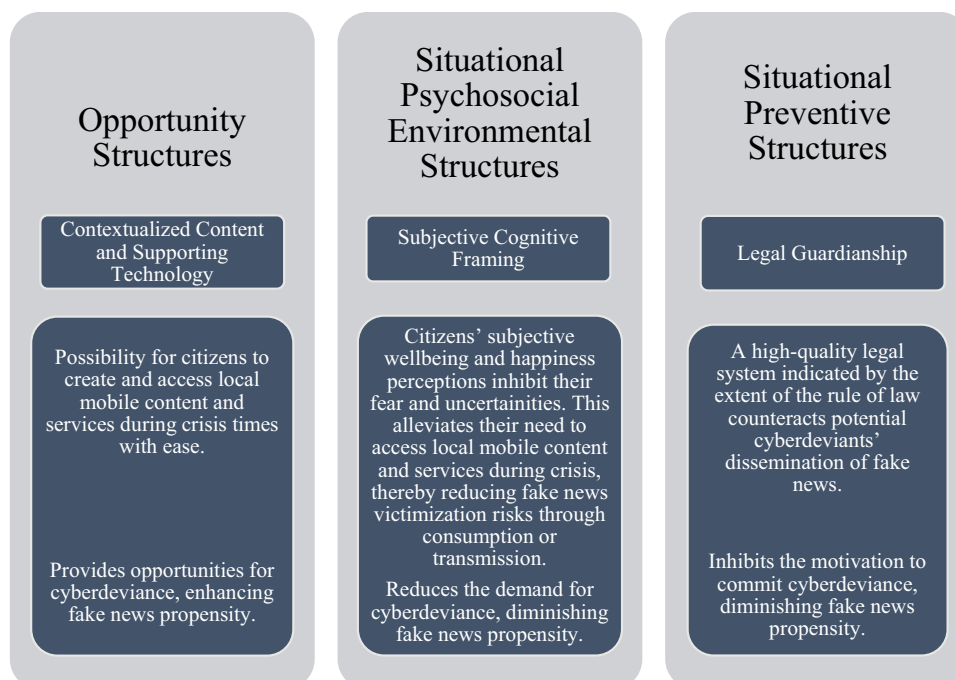


Figure 1. Situational crime prevention analysis: fake news propensity during a crisis.

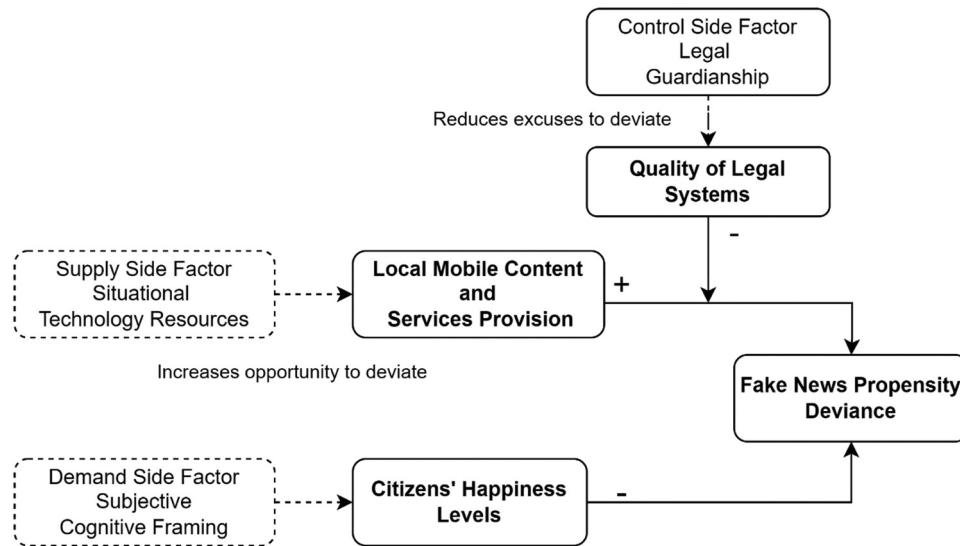


Figure 2. Research model explaining fake news propensity at the country level.

generate and disseminate content that appeals to local populations' contextualized emotions and sentiments (Chaudhuri et al., 2024; Naidoo, 2020). Although the security dimension is critical in severe cybercrimes that manipulate technological hardware or software to reach targets, it is less salient in the context of fake news propensity. In this case, the availability of local mobile content and services alone emerges as a significant proximal technological factor.

In scenarios where the local population extensively uses mobile phones to access services in regional languages, deviant actors gain opportunities to disseminate culturally tailored fake news. Such misinformation is often more believable for local audiences, increasing its likelihood of dissemination. This is particularly relevant during crises, such as the coronavirus pandemic, because during these times, local populations seek geographically contextualized solutions. A greater availability of local mobile content and services provides increased opportunities for cyberdeviant actors, thereby amplifying the volume of deviance (Cook, 1986). Consequently, the availability of technological connectivity, particularly through localized mobile content and services, is associated with greater opportunities for producing and disseminating fake news. Accordingly, we hypothesize:

H1: *The level of local mobile content and services available in a country is positively associated with the level of fake news propensity in that nation.*

3.2. Linking citizens' subjective well-being and happiness to fake news propensity

Cybercrime frequently exploits negative affective states such as fear and uncertainty, increasing individuals' vulnerability to fake news victimization (Naidoo, 2020). Deviant actors use these emotions in

social engineering attacks, while citizens' heightened reliance on information technologies for reactive coping during crises further amplifies susceptibility to cybervictimization (Cheng et al., 2020; Shirish et al., 2021). During the Covid-19 pandemic, cybercriminals exploited these conditions by disseminating unverified information to digitally connected users (Naidoo, 2020; Newman et al., 2020). Together, this evidence indicates that elevated general strain and negative affect increase societal vulnerability to cyberdeviance, including fake news (Naidoo, 2020).

By contrast, higher societal well-being reduces the likelihood of cyber victimization (Helliwell et al., 2021). Greater life satisfaction is associated with lower victimization risk due to reduced general strain and less reactive coping (Estévez et al., 2009; Moore et al., 2012; Olson et al., 2021). Positive subjective well-being is also linked to protective resources such as resilience, optimism, problem-solving ability, and reduced fear of crime (Alfaro-Beracoechea et al., 2018; Baumgardner & Crothers, 2009; Fredrickson & Joiner, 2002; Jo et al., 2022). Consequently, citizens with higher happiness and life satisfaction are less likely to become targets of cyberdeviance, including fake news.

At the national level, happiness functions as a systemic form of subjective cognitive framing that shapes citizens' responses to crises and reduces vulnerability to fake news. Collective well-being captures population-level subjective perceptions and provides insight into societal psychological conditions during periods of uncertainty, such as the onset of the Covid-19 pandemic (Adam et al., 2020; Helliwell et al., 2021). Prior research shows that happiness is a significant protective factor against both crime and victimization (Nikolic-Ristanovic, 2014; Olson et al., 2021). For example, Bhutan—characterized by high reported happiness—also exhibits low crime victimization

rates (Nikolic-Ristanovic, 2014). Conversely, victimization is associated with negative emotional states, including anger, sadness, and depression, whereas positive emotions reduce victimization risk (Tella et al., 2008). Crime and deviance further impose substantial psychological costs on victims, underscoring the importance of preventive affective conditions (Mahuteau & Zhu, 2016; Møller, 2005; Powdthavee, 2005). During crises, positive subjective framing fostered by national happiness thus operates as a critical demand-side situational factor that reduces victimization risk (Shirish & Kotwal, 2023; Shirish et al., 2021).

We argue that positive subjective cognitive framing—operationalized through national subjective well-being and happiness—curbs the spread of fake news by fostering communal resilience and reducing citizens' vulnerability to cyberdeviance (Cook, 1986). Fake news during crises is embedded in both the information environment and social actors, making its mitigation a “whole-of-society” challenge (Gradoń et al., 2021). By reducing susceptibility and demand for misleading information, higher national happiness levels constrain opportunities for fake news transmission (Nikolic-Ristanovic, 2014). Consistent with opportunity theories of crime and deviance, reduced demand lowers the availability of suitable targets and the volume of deviance (Cook, 1986). Accordingly, higher national happiness is associated with a lower risk of victimization during crises such as the Covid-19 pandemic. Based on this reasoning, we hypothesize:

H2: *Citizens' subjective well-being and happiness are negatively associated with the level of fake news propensity in that nation.*

3.3. The role of legal systems in reducing the propensity to deviate

Fake news propagation is a form of deviant behaviour that shares similarities with folk crimes, that are often rationalized under the assumption that such actions are widespread and socially acceptable (Clarke et al., 1976; Freilich & LaFree, 2017). To deter such behaviours, an effective legal system is often necessary. Prior research emphasizes that a high-quality legal system mitigates cyberdeviance (Dashora, 2011). Similarly, studies on curbing fake news dissemination highlight the critical importance of legal systems in addressing this issue (Renda, 2018).

The quality of a country's legal system is reflected in the prevailing rule of law, defined as the extent to which citizens' rights are protected and law enforcement authorities operate efficiently and impartially. Prior research indicates that public trust and legitimacy, fostered by a high-quality legal system, significantly influences the moral fabric of the society,

encouraging moral responsibility in citizens' behaviours (Tyler, 1997). One established approach to crime prevention involves fostering a moral and ethical commitment to the rule of law (Levy et al., 2014). Stronger perceptions of the rule of law make individuals more likely to comply with laws because they have an underlying trust in the judicial process. Perceived legitimacy arises when members of society, including potential deviant actors, regard authorities as having earned the right to command, thereby instilling an obligation to obey (Levy et al., 2014).

The impartial enforcement of laws, a core principle of the rule of law, prevents private and political actors involved in the supply of fake news from circumventing societal legal norms and regulations. The rule of law increases the perceived cost of deviance for potential offenders by ensuring that laws are transparent, enabling deviant actors to understand the consequences of their actions before they engage in behaviours related to the creation or transmission of fake news. According to situational deviance prevention analysis, the presence of legal guardianship provides societal safeguards against deviance (Freilich & Newman, 2017). When the rule of law is upheld through a strong judiciary and legitimate legal institutions, there is less scope for moral dilemmas, and individuals are discouraged from rationalizing deviant acts. This framework also diminishes the perceived rewards of engaging in cyberdeviant behaviours such as the propagation of fake news. Therefore, a strong rule of law acts as a deviance-reducing legal structure, negatively influencing the supply-side dynamics of fake news propensity (Freilich & Newman, 2017). Based on this reasoning, we posit that a strong legal system reduces the likelihood that potential offenders will formulate a rationale for engaging in fake news propagation.

H3: *The strength of the relationship between local mobile content and services and the level of fake news propensity in a country is negatively moderated by the quality of that nation's legal system indicated by the extent of the rule of law in that country.*

4. Research design: sequential mixed method

To investigate the antecedents and boundary conditions of a potential theory explaining fake news propensity at the national level, we employed a sequential mixed methods design combining a dominant quantitative strand with a less dominant qualitative component. This research design allows for both generalizability through statistical testing, and depth through interpretive insights, thereby facilitating robust meta-inferences (Venkatesh et al., 2013). In

the first phase, we tested a theorized model using secondary cross-national datasets to examine the relationships between local mobile content and services, subjective well-being, and fake news susceptibility, while also controlling for relevant national characteristics. Building on the statistical findings, the second phase comprised a qualitative study with expert policy actors to assess the contextual and practical relevance of our theorized model, and identify its potential limitations and boundary conditions.

This sequential mixed methods design served three purposes: (1) to validate the theoretical model developed through quantitative testing, (2) to triangulate and refine the initial findings through contextual expert narratives, and (3) to integrate insights into a comprehensive framework with theoretical and policy implications (Busse et al., 2017; Matook et al., 2024; Shirish et al., 2023). By integrating both strands, our study ensures methodological rigour and practical resonance, particularly in understanding the dynamics of misinformation propagation during critical socio-political crises such as the Coronavirus pandemic.

5. Quantitative study: data

Our quantitative empirical analysis drew on data from seven primary sources: (1) the *Coronavirus Infodemic Database* by the International Fact-Checking Network (IFCN) of the Poynter Institute 2020; (2) the Global System for Mobile Communications Association (GSMA) Foundation's *Intelligence Report 2020*; (3) the *Heritage Foundation Report 2020*; (4) the *Freedom in the World Index Report 2020*; (5) the *Reporters Without Borders Report 2020*; (6) the *Global Health Security Index 2020*; and (7) the *World Happiness Report 2020*, which is produced by Gallup, the United Nations Sustainable Development Solution Network, and Oxford Wellbeing Research Center. To ensure consistency, we followed a methodological approach used in prior studies, including only those countries for which data were available in all seven reports (Shirish & Kotwal, 2023; Srivastava et al., 2016). This criterion resulted in a final sample of 65 countries; these are listed in Table 1.

5.1. Constructs and measures

Full details of all the variables used in this study, along with their sources, are provided in Appendix A. We

also provide some of the salient details in the description below.

5.1.1. Dependent variable

We measured our dependent variable, *fake news propensity*, using the volume of Covid-19-related fake news originating in different countries during the early months of the pandemic, as recorded in the Coronavirus Infodemic IFCN Database. Other studies have also used the IFCN database to analyze Covid-19-related fake news phenomena (Brennen et al., 2020; Humprecht et al., 2021; Shirish et al., 2021). We systematically collected data on the volume of Covid-19-related fake news across 65 countries (Table 1) for the period between the onset of the pandemic and June 2020. This timeframe marked a peak in the emergence and spread of coronavirus-related fake news, leading to the United Nations' coining of the term "infodemic" (Poynter, 2020). Because IFCN is recognized as a well-established global body working in partnership with several organizations, the data from its database can be deemed reliable and valid for measuring fake news propensity, and they have been used in prior studies such as Shirish et al. (2021).

5.1.2. Independent variables

The independent variables for this study—that is, the situational opportunity structure, local mobile content and services, and the situational psychosocial environmental structure, *citizens' subjective well-being and happiness*—were derived from cross-country databases, details of which are provided in Appendix A. *Local mobile content and services* were measured using the mobile connectivity index provided in the GSMA Intelligence Report 2020. *Citizens' subjective well-being and happiness* were based on data from the *World Happiness Report 2020*. Both agencies are widely recognized for their expertise and have over a decade of experience in their respective domains. We measured the moderator variable, the *rule of law*, which is derived from the World Freedom Report 2020 and captures the preventive role of legal guardianship. This measure evaluates factors such as judicial independence, equality before the law, adherence to due process, protection against illegitimate physical force, and freedom from war and insurgencies.

The data for all research variables were derived from either complex factual datasets or large-scale global public opinion surveys administered by the

Table 1. List of countries analyzed.

Argentina, Australia, Bangladesh, Belgium, Bolivia, Brazil, Burundi, Cameroon, Canada, Chile, China, Colombia, Côte d'Ivoire, Croatia, Denmark, Egypt, Finland, France, Germany, Ghana, Greece, Guatemala, India, Indonesia, Iran, Ireland, Italy, Japan, Jordan, Kazakhstan, Lithuania, Madagascar, Malaysia, Mali, Mexico, Morocco, Netherlands, Nigeria, Pakistan, Paraguay, Peru, Philippines, Poland, Portugal, Russia, Rwanda, Saudi Arabia, Senegal, Singapore, Slovakia, South Africa, South Korea, Spain, Sri Lanka, Sweden, Tanzania, Thailand, Tunisia, Turkey, Ukraine, United Arab Emirates, United Kingdom, United States, Uruguay, Zimbabwe.

Total Number of Countries Analyzed = 65

forementioned reputable agencies. These organizations employ rigorous and transparent procedures—including standardized measurement instruments, expert-coded assessments, and multi-stage validation protocols—to ensure the reliability, validity, and cross-national comparability of the data collected. While some indicators—particularly those related to citizens' subjective well-being—are perception-based at the individual level, the measures used in this study represent aggregated country-level indices derived from large samples and consistent data collection frameworks, thereby reducing sensitivity to short-term fluctuations in public sentiment. Likewise, institutional variables such as the rule of law are grounded in expert-rated evaluations of enduring legal and governance structures, which evolve gradually over time rather than changing episodically. Consistent with prior cross-country studies, data from these agencies are widely regarded as suitable for conducting rigorous comparative analyses of structural and contextual phenomena (Shirish et al., 2021; Srivastava et al., 2016) and have been utilized extensively in recent academic research, further attesting to their construct stability and analytical suitability (e.g., Shirish & Kotwal, 2023; Shirish et al., 2021).

5.1.3. Control variables

A nation's Covid-19-related fake news propensity may have been influenced by factors beyond those hypothesized in our research model. To ensure the robustness of our results, we carefully selected appropriate country-level control variables by (1) reviewing prior country-level studies, and (2) identifying factors likely to influence citizens' uncertainties and fears during the Covid-19 pandemic, which could, in turn, affect fake news propensity. Five categories of control variables were identified for this study: *the economy*, *citizens' freedom perceptions*, *technology*, *health*, and *education*. Details on all variables are provided in Appendix A. We likewise, obtained our data for the control variables from reliable, independent agencies, including the Global Health Security Index, GSMA Intelligence Report, and Heritage Foundation Reports.

- (1) Economy: Three economy-related country-level factors used as controls in our analysis are (a) gross domestic product (GDP), and (b) unemployment rate, as these might be related to citizens' resilience to crises and deviant behaviour.
- (2) Citizens' Freedom Perceptions: Prior research has shown that economic and media freedom decrease fake news propensity, and political freedom increases fake news propensity. Thus, all three of these freedoms are controlled for in the study.

- (3) Technology: We use two national-level technological factors in our analysis. These are: (a) mobile infrastructure, and (b) mobile affordability. These factors lay the foundations of internet connectivity and, therefore, access to the information technology resources that citizens use for reactive coping during crises.
- (4) Health: We used access to quality health care as a control for health-related uncertainties; the fear these created among citizens may have caused them to fall victim to fake news when they engaged in reactive coping via internet connectivity.
- (5) Education: A country's adult literacy rate might determine access to and the ability to discern fake news. Hence, it was considered a suitable control variable in our research model.

6. Quantitative study: analysis and results

Appendices B and C present the descriptive statistics and correlations for all variables used in our study. Since our dependent variable (fake news propensity) is a count variable, we employed negative binomial regression to test our hypotheses. Negative binomial regression is a generalization of Poisson regression, which is often used to model count data. It loosens the restrictive assumption made by the Poisson model that the variance is equal to the mean (Hilbe, 2011). As Appendix B shows, fake news propensity is an over-dispersed variable, with a variance exceeding the mean. Therefore, we deemed negative binomial regression the most appropriate analytical method for this study (Chaudhry et al., 2020; Medappa & Srivastava, 2020; Shirish et al., 2021). To address potential concerns about multicollinearity, we conducted a variance inflation factor (VIF) test. All computed VIFs fell below the recommended threshold, suggesting that multicollinearity does not confound our results (Davis et al., 2009; Hair et al., 2006; James et al., 2019).

Table 2 presents the results of the regression analyses examining the influence of local mobile content and services and citizens' subjective well-being and happiness on fake news propensity. Model 1 includes the control variables (GDP, literacy, unemployment rate, access to quality healthcare, political freedom, economic freedom, media freedom, mobile infrastructure, and mobile affordability). Model 2 introduces independent variables to isolate their effects. Finally, Model 3 incorporates the moderator rule of law and the interaction term between it and local mobile content and services.

Results from Model 2 indicate that the situational opportunity structure variable local mobile content and services has a positive and significant relationship with fake news propensity ($\beta = 0.029$, $SE = 0.011$, $p = 0.000$),

Table 2. Results of negative binomial regression on fake news propensity.

Var	Model 1 (Controls Only)			Model 2 (Main Effects)			Model 3 (Moderation Effects)		
	(β)	Std Error	Wald Chi-Square	(β)	Std Error	Wald Chi-Square	(β)	Std Error	Wald Chi-Square
Intercept	1.803	0.903	3.985	4.051**	1.989	4.1482	5.898**	1.623	13.203
CON									
GDP	7.05E-5	4.39E-5	2.581	5.58E-5	5.104E-5	1.193	6.14E-5	4.57E-5	1.802
LIT	.012	.011	1.170	.013	.015	.723	0.11	.011	1.029
UNE	-.006	.030	.046	-.004	.040	.009	-.034	.031	1.242
QHC	-.006	.009	.367	-.004	.012	.079	-.002	.001	.027
PFD	.472**	.110	18.423	.379**	.111	10.137	.314**	.111	7.937
EFD	-.587**	.164	12.886	-.447**	.148	9.137	-.497**	.171	8.447
MFD	-.470**	.185	6.440	-.429*	.172	6.182	-.343	.192	3.192
MIF	.003	.014	.039	-.029	.0260	1.285	-.040	.022	3.391
MAF	.037**	.014	7.226	.029	.0180	2.508	.021	.016	1.796
MEV									
MCS				.029**	.011	7.366	.019*	.008	5.319
HAP				-.581**	.226	6.584	-.274	.223	1.508
MOD									
ROL							.107*	.043	6.166
MCS*ROL							-.004**	.001	14.099
Df	55			53			51		
L-L	-332.576			-328.191			-323.763		
L-L CS	53.968			62.736			71.593		
MS	.000			.000			.000		
AIC	685.151			680.383 [^]			675.526 [^]		
BIC	706.895			706.476			705.968 [^]		
N	65			65			65		

Var: Variable Names; CON: Control Variables; MEV: Main Effects Variables; MOD: Moderating Variables.

INT: Intercept; GDP: Overall Gross Domestic Product; LIT: Literacy; UNE: Unemployment; QHC: Access to Quality Health Care; PFD: Political Freedom; EFD: Economic Freedom; MFD: Media Freedom; MIF: Mobile Infrastructure; MAF: Mobile Affordability; MCS: Mobile Content and Services; HAP: Happiness; ROL: Rule of Law.

Model Indicators:

* $p < 0.05$; ** $p < 0.01$.

Df: Residual degrees of freedom of the model; L-L: Log Likelihood; L-L CS: Log Likelihood Chi-Square; MS: Model Significance; AIC: Akaike's information criterion; BIC: Bayesian Information Criterion Models 2 and 3 are better-fitting models compared to Model 1.

N: Number of observations used in the calculation.

supporting H1. Similarly, the situational psychosocial environmental structure variable, citizens' subjective well-being and happiness, has a significant negative relationship with fake news propensity ($\beta = -0.581$, $SE = 0.226$, $p = 0.006$), supporting H2. Model 3 shows that the situational preventive structure variable, rule of law, moderates the relationship between local mobile content and services and fake news propensity, reducing the independent variable's impact ($\beta = -0.004$, $SE = 0.001$, $p < 0.000$). Therefore, we also find support for H3. The Akaike information criterion (AIC) and Bayesian information criterion (BIC) values indicate model fit. Both metrics are lower for Model 2 than for Model 1, and lower still for Model 3 than for Model 2, suggesting that Model 3 offers the best fit. Among the control variables, only political freedom and economic freedom exhibit significant positive and negative relationships, respectively, with fake news propensity. None of the other control variables significantly influences fake news propensity in the full model.

7. Qualitative study methodology

The qualitative strand was designed to complement and extend our quantitative analysis by providing nuanced, context-specific insights into the phenomenon of fake news propensity. To this end, we conducted in-depth, semi-structured interviews with eight experts in public policy, civil society, and digital media

regulation, all of whom have direct experience dealing with misinformation and digital governance.

Participants were identified using purposive sampling via professional platforms like LinkedIn, with selection criteria ensuring formative validity (Lee & Hubona, 2009)– each expert possessed substantial expertise in the domains of technology, regulation, and fake news. We initially identified and contacted 12 potential interviewees, of whom eight agreed to participate. Detailed profiles of the interviewees are provided in Table 3. The interviewees' extensive experience with fake news meant they were well-positioned to relate to and interpret the findings holistically.

Prior to the interviews, we shared our research questions and quantitative findings with the interviewees to gather their views on our theorized research model. During the interviews, we discussed the model and posed questions regarding the applicability of the results to policy and practice. We also asked them about potential boundary conditions or scenarios in which the model might not be applicable. Additionally, interviewees were encouraged to offer constructive critiques of our quantitative findings. We conducted eight purposively sampled expert interviews, each lasting between 60 and 90 minutes, via telephone or videoconference. Six were transcribed from recordings (MS Teams or Zoom), and two were based on detailed notes due to poor audio quality. The resulting dataset consisted of approximately 260

Table 3. Details on Expert interviewees.

Experts	Role	Domain Expertise	Category
Expert A	Marketing and consumer psychology and Policy Expert, High-Level Executive, Government of India	Press and Information Communication Policy	Government Services
Expert B	Senior Journalist, AFP	Media and Journalism	Professional Media Agency
Expert C	Academic Expert, Member of Special Expert Task Force on Disinformation at The French Regulatory Authority for Audiovisual and Digital Communication (ARCOM)	Academia and Policy Making	Government Advisory
Expert D	Activist for the Disinformation Project in a US state	Social activists and media	Social Activism
Expert E	Advisor for Communication and Information, UNESCO	Communication and information policy	International Body, UN.
Expert F	Director, Social and Regional Research at GSMA Intelligence	Mobile Technology Policy and Fake News	Global organization unifying mobile ecosystem
Expert G	Senior Research Analyst	Evidence-based policymaking to counter threats within the information environment	Global Organization Think Tank
Expert H	Political Scientist/Policy Expert	Misinformation and Disinformation Policy	European Union Public Diplomacy/Editor

pages of single-spaced text, forming a rich basis for qualitative analysis.

Although the sample size is modest, it aligns with recognized qualitative research norms, especially in expert and theory-validating studies. Specifically, Fugard and Potts (2015) propose a quantitative tool for determining thematic sample size that takes into account (a) the expected prevalence of themes, (b) the number of instances required, and (c) analytical power. Their model demonstrates that for themes of moderate prevalence (e.g., 20%), a relatively small number of interviews can still yield sufficient insight (Fugard & Potts, 2015). In our study, the six initial interviews reached thematic saturation, and two additional interviews confirmed that no new themes emerged, indicating confidence in our theme coverage (Guest et al., 2006). Given the in-depth nature of conversation with high-expertise participants and convergence in emergent themes, we concluded that eight interviews offered an information-rich and methodologically sound sample for exploring contextual boundary conditions and validating our cross-country quantitative model.

We applied thematic analysis to examine the expert interview data, using an iterative and inductive coding process tailored to our exploratory aims. The qualitative strand aimed to identify contextual boundary conditions and interpretive insights that complement and validate the cross-country quantitative model (Braun & Clarke, 2006). By integrating the insights obtained in the qualitative study with our quantitative findings, we validated the theorized model (confirmatory purpose of the mixed methods design). We also delineated boundary conditions of the theoretical model and derived practical recommendations, thereby enhancing the transferability and contextual validity of the study's implications (complementary purposes of mixed method design) (Matook et al., 2024; Srivastava & Chandra, 2018). In this section, we detail the data collection methods of our qualitative

study, while in the next section, we present our analysis and the metainferences from the mixed methods. Metainferences can be represented using various forms, including diagrams (Shirish et al., 2023; Venkatesh et al., 2013). In this study, we discuss and present our metainferences via Figure 3 and further report on practical implications for both practitioners and policymakers (Matook et al., 2024; Shirish et al., 2023).

8. Qualitative study: analysis, findings, and metainferences

8.1. Qualitative findings: establishing credibility

Following Braun and Clarke (2006) thematic approach, we began with open coding of interview transcripts to capture initial concepts. Coding was refined through multiple iterative rounds, using constant comparison within and across interviews to explore patterns and variations in experts' perspectives on the phenomenon of fake news propensity across diverse geopolitical and regulatory contexts. Throughout data collection, we maintained reflexive notes immediately after each interview to document early impressions, contextual nuances, emerging codes, and analytic questions. These notes informed and enriched the evolving coding process, supporting transparency and analytic depth. Our analysis focused on identifying two key types of patterns: (1) emergent contextual conditions, and (2) policy-relevant insights or solution pathways proposed by experts. To enhance credibility, all themes were firmly grounded in the data and are supported by illustrative quotes in the findings section. We also conducted team-based coding involving a multidisciplinary author group, including scholars from communication, law, and management, which allowed for investigator triangulation and critical dialogue around theme development. This collaborative, reflexive, and data-

grounded approach ensured that our interpretations remained well-substantiated, contextually sensitive, and credible in light of the study's theoretical and policy-oriented aims.

Consistent with our research design, the interviewees endorsed the credibility of our theorized model (Miles et al., 2020). They emphasized that access to local mobile content and services, as well as citizens' subjective well-being and happiness, are significant factors influencing the cyberdeviant behaviour of fake news propagation. Additionally, experts supported the moderating role of high-quality legal systems in mitigating the negative impact of local mobile content and services on fake news propagation. All interviewees acknowledged the relevance of the theoretically identified variables in shaping fake news propensity within nations. Two expert comments particularly highlight key aspects of the research model:

Digital technology is at the center of how society functions ... locally relevant content and services are enablers of mobile internet and digital services adoption. However, **access to more local content**² can accelerate the prevalence of fake news ... misinformation is normally spread not by locally produced platform owners/services but by users of these local platforms that create and spread fake news ... **legal systems should be robust** enough to tackle fake news as well. I do believe that **happy citizens** are less likely to spread fake news; nevertheless, I can imagine that such a person can also become a victim if he is not critical of the news he/she consumes via social media platforms ... without targeted education on how to consume online content, fake news can continue to spread. *Expert F*

More than detection, elimination of **bad local content and services** from local content platforms, we need to balance them with a human-centric approach to stop the spread of fake news ... increasing the happiness **levels** and quality of lives are important, so more focus should be on people and their relationship to news and autonomy. **Local content and services** need to be better regulated to stop the spread of fake news ... having a quality legal system that is enforced at the local levels should go hand in hand in this effort to mitigate the spread of fake news via local mobile content and services in a country. *Expert D*

These expert confirmations lend practical credibility to our theoretically driven and empirically tested research model (Figure 2).

8.2. Unearthing boundary conditions and proposing policy insights

Although the interviewed experts affirmed the salience of all the theorized variables and relationships in our research model (Figure 2), it was essential to identify critical boundary conditions—i.e., factors that may

limit the model's applicability. Understanding these conditions contributes to a theoretical understanding of the phenomenon that is both holistic and grounded in real-world contexts. Additionally, identifying boundary conditions offers valuable directions for future research and actionable policy insights to facilitate the implementation of practical initiatives that are based on our theoretical findings (Shirish et al., 2023; Venkatesh et al., 2013). Our analysis of the interview data identified three critical boundary conditions that influence the salience of the theorized variables. We now present these boundary conditions along with related policy insights that emerged from the analysis. These findings represent significant additions to our results and may serve as a foundation for future research exploring fake news propensity across nations (Shirish et al., 2023).

8.2.1. Boundary condition #1: presence of integrated digital literacy programs

The experts whom we interviewed emphasized the need to raise public awareness by enhancing citizens' digital literacy so they can identify fake news. Governments are increasingly investing in digital literacy education to improve local mobile content and services, aiming for a positive local impact. However, experts noted that many of these programmes lack a targeted focus on fake news prevention, detection, and management. Integrating fake news-related education into such initiatives is seen as essential for addressing this gap. One expert highlighted the need for holistic training programmes to restrict the supply-side dynamics of fake news propensity:

"**Digital skills** (upliftment) is one of the crucial pillars to strengthen local mobile content and services provisions ... Rwanda's government has an excellent program based on digital skill ambassadors; for example ... since digital literacy programs are not uniform (content or frequency).(yet) one thing they can all do is to include **fake news-related awareness and education** ... your findings and the call to integrate fake news education with digital literacy programs can attract much attention. They will be good for stakeholders to imbibe. I have seen digital literacy programs centered around teaching people how to log into Facebook, make friends on Facebook, and post pictures on Facebook. However, suppose they (Government-led agencies, international development partners, and civil society) can include training on recognizing fake news and how not to reshare any content in those modules. In that case, they are not confident of the veracity of how to report malicious information intended to cause harm, which would be a very good thing. So yes, it is important that digital skills go **beyond just teaching the how of online**, but also include the what ... because it is one thing to know how to go online and use digital technologies and the

mobile Internet. Still, it is also something else to **know what to access, what to read, what to accept, what to share**, and things like that". - *Expert F*

Moreover, the experts expressed the belief that such human-centered approaches are superior to technical solutions when it comes to preventing and detecting fake news propagation.

"You can spend so much time and so many resources trying to Implement these like **tech-based** solutions where you recognize fake news, and you detect it, and you mark it as such, and you are eliminating bots from platforms, etc to some extent these are useful, but **it is never going to work** if it is not partnered with and there's not a balance of these **human-centered approaches** (training programs) too, because at the end of the day, it really comes out to the person that's reading this stuff and what their intentions are and what their perspective is and whether they are able to or willing to think critically about these things So I think these results really tie into my research to focus more on people and their relationship with news and autonomy over information consumption". - *Expert D*

Associated Policy Insight #1: public financing and leveraging participation for educational policy initiatives. One key policy insight linked to this boundary condition is the need to leverage participation from diverse stakeholders, including government agencies, international funding bodies, civil society groups, and media organizations, to collaboratively design and implement public policy innovations. For example, integrating fake news management modules into funded digital literacy programmes, such as those supported by European projects, can empower citizens to identify and prevent fake news. Policy targets for this strategy include regional funding agencies, civil society groups, and media. This strategy impacts the agenda-setting stage as well as the design and implementation stages of a given policy lifecycle. One expert mentioned that identifying and preventing fake news and **raising awareness among citizens** should happen at different levels of the government and within civil society:

"...allowing the federal government or the federal government allowing the local government or local institutions also to have a big role in this (prevention of fake news) would be huge. They also know the local content providers better. They know their populations better. That is one thing with the disinformation project we are also focused on I recommend the government integrate local populations (media providers) and have local media influencers to **raise awareness** towards **preventing fake news** and **propagation of right news, especially during crisis times**". - *Expert D*.

Furthermore, allocating public funding to print media could complement educational initiatives aimed at preventing fake news. Consequently, it is important to support and finance local independent journalism, including

community-embedded newspapers and investigative reporting, as a counterbalance to fake news.³ One media expert underscored the importance of this approach:

"Two important policies to consider are **public financing** for the printing press to better prepare for future crisis-related fake news rise and **long-term education** specifically focusing on reading the news and how to discern fake from real news". - *Expert B*

8.2.2. Boundary condition #2: upholding democratic values

Several democratic values must be upheld and carefully balanced when addressing the propensity for fake news production and dissemination. Among these, we identify *equality* as a foundational democratic value. The absence of equality can diminish happiness levels, counteracting the scope of the latter to reduce fake news propensity. Additionally, inequality heightens citizens' victimization risk, thereby influencing the demand-side drivers of fake news by increasing their susceptibility to misinformation. Inequality perceptions may also provide motivations for deviant behaviour, such as the production and dissemination of fake news. This, in turn, amplifies the supply-side dynamics of fake news propensity by fostering a conducive environment for its creation and spread.

The perception of equality within a nation influences the cognitive mindset of its citizens, thereby reducing victimization risks. In countries with significant disparities between the rich and the poor, civil unrest often emerges. Such unrest reflects a lack of healthy cognitive framing and may provoke deviant behaviour. This highlights the importance of addressing inequality as part of policy agenda-setting to foster societal well-being and reduce the risks associated with fake news propagation. One expert emphasized the critical role of happiness and well-being in government policy:

"We live in an uncertain world with war, massive risk of disinformation, climate change, social unrest, and the risk to democracy, and finally a high risk to happiness. This study clearly shows the need for **governments to find ways to reduce the risk to happiness**. Happiness and well-being of citizens is the key to complex crisis management situations that governments face . . . in my experience, politicians and governments do not have this issue in their priority list, although they must". - *Expert C*

Another expert highlighted that ensuring equal access to government facilities, services, and information is essential for fostering happiness and reducing the likelihood of deviant behaviour:

"Happiest societies are the most equal ones. If people feel they (rich and poor) have **equal access to all government facilities** and services, then fake news will not find a foothold in such societies . . ." - *Expert B*.

Moreover, the perception of equality influences the link between the quality of the legal system and the supply-side dynamics of fake news. Fairness and predictability in legal norms and their enforcement, grounded in principles of equality, play a critical role in addressing fake news propagation. Discussing this point in depth, one expert commented:

“...it is not too difficult for governments to identify groups of people habitually indulging in creating and spreading fake news who offer local language content through mobile devices for political or economic reasons So when the general public is confident that the **government will be fair** in how it views, policies, and punishes fake news perpetrators, keeping in view **principles of equality**, it will certainly make a huge impact in reducing the excuses to be involved in such deviance in the local context, at least. Be it a politician spreading fake news for political leverage or an unemployed person trying to attract web traffic for their website for monetary purposes during a crisis situation”. - *Expert B*.

In addition, the experts emphasized the need to regulate local mobile content and services while ensuring diversity and equal access to information sources to address the challenges posed by fake news. News originating from a single or anonymous source is often perceived as dubious due to the lack of corroborating evidence. In contrast, information triangulated from multiple sources fosters greater trust and credibility. Regions where diverse informational sources are widely available exhibit lower levels of fake news propagation, even in contexts of high social media penetration. This speaks to the importance of systematically designing multiple sources of information and ensuring their accessibility to citizens.⁴ Lack of access to multiple information sources constitutes a significant aspect of inequality. Citizens rely on diverse sources to confirm the veracity of information, a critical factor in curbing the spread of fake news. However, unequal access to information can occur both across countries, due to variations in governance structures and support for information intermediaries, and within nations, where systemic barriers such as socioeconomic status or geographic location limit access. For instance, some citizens may have better information resources or the capacity to obtain such resources, while others are constrained by their social milieu or other structural inequalities that limit their opportunity to be both informed and to be heard. One expert elaborated on these disparities:

“If you have a **society that can access information from multiple sources, then people in those countries will have a lesser propensity to spread misinformation** in countries where access to information is limited, then any information you have access to (via social media) becomes the only information . . . therefore we need to make the distinction between mis/disinformation when it comes to regulation . . . In countries such as the US, despite

a high level of education, access to information, and diverse sources, we still find fake news. This is because they are mostly disinformation and not misinformation that is prevalent in such countries . . . here we need to regulate and punish the criminals strictly”- *Expert F*

Another expert echoed the lack of citizens' voice as a fundamental structural barrier that contributes to the unhappiness levels of citizens, thus making them fall prey to fake news easily.

... you could tackle the problem of misinformation by trying to, of course, empower people to detect it—to not be manipulated—by stopping the flow. But also, there is a problem that is, you know, the root cause of all that, which is what we were talking about earlier: the unhappiness of citizens . . . **as long as your citizens, your constituents, are not happy or are not feeling heard, there's going to be continued opportunities for manipulation** - *Expert H*

The second democratic value identified is *freedom of expression*, which must be carefully balanced with legal and regulatory measures to address fake news. Our expert interviewees emphasized the need to contextualize this study's findings using each country's unique circumstances when implementing strategies to regulate supply-side factors or curb the proliferation of fake news. Discussing these perspectives, two interview respondents highlighted the complexities of balancing freedom of expression with regulatory efforts:

“**Context matters**, more control of local content and service provision and losing up on external content and service provision can be concerning . . . **strategies taken toward deviance regulation can sometimes have a disproportionate role and indirectly impact policies taken by governments in other sectors**, it can also curtail public freedom . . . so I am skeptical about regulations as well as any control of local content and service provisions however this issue is **specific to each country context** . . . improving the engagement with citizens, **developing social connections, empathy and strengthening well-being** are better approaches to fake news prevention”. - *Expert E*

“Governments need to know where to draw the line between **attempts to stop fake news** (disinformation) and their attempts to **infringe freedom of expression**”- *Expert F*

Associated Policy Insight #2: Enhancing Rigor and Relevance of Regulation Via Both Soft and Hard Approaches. This policy insight highlights the importance of fostering a healthy cognitive mindset among citizens at the national level through well-being-based strategies to mitigate recidivism risks. Additionally, it advocates for the strategic use of legal frameworks to regulate local mobile content service providers and confront the actions of fake news perpetrators. The proposed strategies target two groups: local content and service providers on the one hand, and potential

deviant actors responsible for generating and disseminating fake news on the other. These contextualized approaches influence both the agenda-setting and implementation stages of the public policy lifecycle, encompassing the detection, prevention, and management of fake news deviance.

After the law has punished perpetrators for acts such as fake news production and dissemination, they may still be exposed to the criminogenic factors that drove them to commit those acts. To address this issue, a multipronged approach is required, combining the detection of local deviant actors with the application of targeted legal provisions through both hard and soft approaches.

A human-centered approach, including well-being training to reduce the risk of recidivism, is proposed by experts as a soft-touch strategy for governments and policymakers. By fostering a kind of “inoculation” against fake news, governments can help to curb misinformation and protect citizens from it. Our policy experts emphasized that creating a positive psychosocial environment, rooted in citizens’ subjective well-being, can serve as a preventive mechanism against fake news. In a happy and supportive society, citizens are less likely to consume or promote fake news, as a positive collective climate enhances their cognitive resilience. Our expert interviewees also stressed that governments worldwide must prioritize citizen well-being as a long-term strategy for handling fake news. Enhancing well-being-led institutions and offering targeted training can mitigate recidivism risks and reduce opportunities for fake news production.

“As a long-term strategy, every country has to increase its well-being to prevent opportunities to engage in deviance; it is essential. It will also be useful to have a multiprong crisis strategy that starts with first identifying the community or group of people perpetrating fake news in the local territory and offering them specific well-being training solutions as a preventive mechanism to be best planned for future crisis situations” – Expert A

“A lot of citizen-level awareness (critical thinking and global citizenship education) needs to be done Being happy from a holistic perspective involves having meaningful relationships (self and with others). If we don’t have a good sounding board and reference groups around us, a lot of things that we see (fake news) may seem plausible. Therefore, having a community connection and diversity of contacts is very important. Countries need to strengthen well-being-led institutions as a long-term strategy (fake news prevention)” – Expert E

“The root cause [of misinformation] is unhappiness of citizens . . . that is why in my professional roles, I have been advocating for participatory and deliberative democracy [that provides chances for citizens’ voices to be heard], because this really goes hand in hand [with other policy responses to the misinformation]” – Expert H

Another soft-touch approach proposed by policy experts involves distinguishing between local service providers (e.g., platforms) and local content providers to better regulate the flow of information on social media platforms. The experts emphasized the importance of fostering collaborations with social media platforms through both regulatory measures and stakeholder engagement initiatives. Doing so would prevent the creation and spread of fake news, including both misinformation and disinformation, within local networks. Additionally, this approach allows governments to focus on the most significant local content providers operating on social media rather than imposing blanket regulations on all content providers. By targeting key actors, this strategy strikes a balance between protecting liberty of expression and regulating fake news, ensuring that regulations are both contextually relevant and proportionate to local needs (Pappas et al., 2023).

One of the technology policy experts discussed the key role played by social media content and service providers in propagating fake news and how this role can be impacted by *information access and regulation*, thus hinting at increasing equality in access to information (Matook et al., 2022; Pappas et al., 2023).

“Usually, local content providers, apart from online/offline news and information agencies, happen via social media. Most content providers are usually regulated (in some form); they can be traced as their websites are nationally hosted and registered. Therefore, the propensity (of such regulated content providers) to disseminate fake news (mis or disinformation) is low. Whereas, if you are a content provider (even a citizen) using a popular social media platform (local or international), you can hide in the guise of anonymity and disseminate mis/disinformation using reports by agencies that provide further details on top use of social media for each country context can provide insights on how to collaborate with social media platforms and regulate fake content providers” – Expert A

Another expert, agreeing with the need for quality legal systems, suggested a more thoughtful tiered approach to regulating fake news. This would aim to reduce victimization risk and opportunities to commit deviance by looking at the role social media technologies play in shaping interactional dynamics, thereby allowing the policy owner adequate discretion to adopt soft or hard approaches to regulating fake news (Matook et al., 2022).

“The quality of the legal system is a very interesting factor to consider. I think that would probably trickle down into quality legal systems at the federal level and then quality legal systems at the local level, so governments can aim to achieve a higher quality of legal system at both levels . . . however, what to regulate and what not to regulate is a tough choice That is why I think allowing more (discretion on what and how to regulate) local governments to regulate, would help I think focusing on the

kind of people (victims/perpetrators) and their relationship with fake news and social media technology is important to understand how to regulate” – *Expert D*

Our findings suggest that citizens are less likely to fall prey to fake news when a nation has a high-quality legal system. By contrast, weak legal systems facilitate the production, dissemination, and consumption of fake news, particularly in uncertain situations, even when citizens are aware of its dangers. These quantitative findings were strongly endorsed by the policy experts we interviewed, who emphasized the urgent need for countries to strengthen their legal systems to mitigate the proliferation of fake news. They highlighted several strategies for building stronger legal systems, including creating stability and predictability, reinforcing judicial ethics and accountability, streamlining court administration, and enhancing legal efficiency and service transparency (Bayer et al., 2019). While underscoring the importance of these measures, the experts also identified two key hard approaches as critical remedies for tackling fake news.

First, our interviewees emphasized the importance of implementing rules and regulations to curb the spread of fake news. One expert highlighted the European Union’s recent adoption of the Digital Services Act (DSA) as a specific regulatory framework addressing fake news. They recommended that countries outside the EU consider integrating principles from the DSA into their legal systems:

“From your results, we note that **regulation can be a good way to decrease the propensity for fake news** . . . The Digital Services Act is one such specific European regulation against fake news and similar offences. It provides ways to reduce the amplification and moderation of fake news in the context of social media use. **Since the quality of legal systems reduces fake news propensity**, more countries can consider adopting similar regulation regimes to improve their regulatory power on such matters, thereby **effectively deterring cyberdeviant actors against creating and spreading fake news**” – *Expert C*

Second, the experts emphasized the need for high-quality legal systems specifically addressing cyberdeviance, including fake news, when deploying cyberservices. The role of regulation, particularly in relation to the algorithmic amplification of misinformation and the influence of foreign actors in setting up disinformation campaigns, is an issue that may necessitate a regulatory approach.⁵

Discussing such legal systems, one expert commented:

“**Information technology and communication-related regulations and laws will indeed reduce the urge to commit cyberdeviances such as fake news.** Most people will think twice before indulging in such

activities when they offer local language content for consumption to the masses . . . In XY (a country in Europe), **several regulations can deter people from spreading fake news targeting local people or companies.** Very often, there are court cases on this matter. It would not be easy in a country without such regulations or laws. People (in those countries) will spread fake news without fear of punishment in such countries”. – *Expert B*

8.2.3. Boundary condition #3: strategically shaping local communication environment

The final boundary condition concerns the communication environment in which fake news thrives. Specifically, the popularity of the local or national language in which mobile content and services are (or can be) delivered is a critical factor in understanding how to combat fake news at a macro level. More specifically, the availability of official local mobile content and services to counter misinformation in local languages is equally important. Our findings suggest that countries where widely spoken languages are prevalent face greater challenges regarding fake news dissemination, particularly during crises. One expert elaborated on this:

“In Nordic countries such as XY (a Nordic country), one might experience less fake news as the number of **people who can offer local mobile content and services is limited** to only those people who can speak the XY language (Nordic language) in the world. XY language is not a widely used language it offers these countries a natural shield against offenders trying to deviate from societal norms for monetary reasons (e.g., virality is limited) it is not the case in countries such as the US, France, India, Spain for example, their national languages are more widely spoken in the world and therefore offering fake content services **becomes lucrative for many offenders**, and consequently it also has more chances of spreading both locally and internationally” – *Expert B*

This observation highlights the need for tailored regulatory strategies, particularly in countries with universally shared languages. At the same time, it is just as important for government actors to provide official news and credible reports in which there is a public interest via local mobile content and services. These findings directly inform the policy insight we share below.

Associated Policy Insight #3: Partnership and Presence of Government in Local Communities to Increase Knowledge Flows. This policy insight emphasizes the need for governments to engage with local communities by promptly and actively promoting official news and credible reports in which there is a public interest. Such policies do not stand in contrast to other policy actions that can enhance citizens’ voices, leading to improved multi-directional knowledge flows (Shirish et al., 2025) and enhancing institutional trust perceptions.

Additionally, partnerships with information intermediaries—such as news providers that prioritize stories whose publication is in the public interest, informed citizen groups, and fact-checking organizations—can enhance the flow of accurate information between governments and citizens (Tambini, 2017). The policy targets would include citizens, local mobile content and service providers, social media platforms, news agencies, and fact-checking bodies. This strategy could involve various stages of the public policy life-cycle, including agenda setting and policy design, monitoring, and implementation. This strategy involves preventing and managing fake news deviance, thus reducing its societal impact in a timely manner.

The need to strengthen governmental presence in local mobile content and services aligns with policy opinions that emerged during the pandemic; these indicate that restructuring national communication environments in a centralized manner can mitigate the spread of disinformation or reduce its impact, thereby enabling societies to address crises using scientifically grounded strategies (Posetti & Bontcheva, 2020). In addition to facilitating access to local mobile content and services, governments should focus on informing and educating the public to promote critical media literacy and responsible information-sharing practices within the digital sphere. The experts we interviewed highlighted the importance of timely and targeted government interventions:

“It may be difficult to completely do away with the provision of local mobile content and services due to its negative externalities, as the study mentions. However, I think countries can take **preventive actions such as increasing their own local social media presence and building clear, timely, relevant, targeted public communications** to counter the impact of these negative externalities during a crisis situation” – *Expert A*

“During crisis periods, such as for health and election periods, we could imagine **more presence of government on social media; we also need to go a step further and partner with platforms and other local content providers to amplify official, right, and relevant information and content. This could be an effective crisis management strategy**”- *Expert C*

These qualitative boundary conditions and the associated policy insights allow us to generate robust metainferences from both qualitative and quantitative studies offering a comprehensive situational analysis to effectively design interventions to prevent of fake news within nations.

9. Metainferences and discussion

Based on our quantitative and qualitative findings and taking cues from foundational work on mixed method research in IS, we present our metainferences via a figure (Shirish et al., 2023; Venkatesh et al., 2013). Figure 3 below, is a synthetic representation obtained from confirming and complementing our initial theorized model presented in Figure 1. This model is self-explanatory and it presents our contextualized theory-driven comprehensive country-level situational crime prevention analysis framework for fake news prevention. Since we have already discussed both the proximal factors that were initially theorized in our research model and have also detailed the uncovered boundary conditions in the preceding section, we move to discuss our findings and enumerate their contribution to the literature and practice.

9.1. Contributions

With the aim of counteracting the significant short- and long-term harms associated with fake news propagation, researchers and policymakers are exploring various

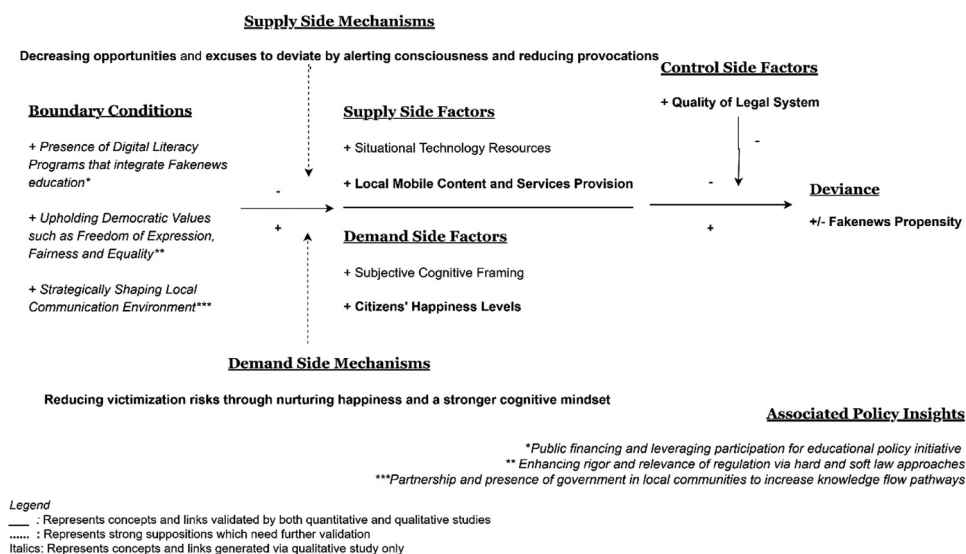


Figure 3. Situational crime prevention analysis framework for fake news prevention.

theoretical frameworks to better understand this socio-technical phenomenon and develop strategies to curb the production and consumption of misinformation (Khuong et al., 2020; Miller et al., 2022; Pennycook et al., 2020; Schoenmueller et al., 2022; Shirish et al., 2021; Wang et al., 2022).

Although prior research has acknowledged fake news as a form of cyberdeviance, the crime-theoretic perspective has yet to be explicitly applied to explain cross-national differences in fake news propensity. Grounded in the IS literature and situational crime prevention analysis, this study contributes to a holistic understanding of the fake news phenomenon in several ways.

First, our research proposes an integrated theoretical framework to explain fake news propensity across nations. We introduce the concept of “supply side” and “demand side” factors—situational opportunity structures and situational psychosocial environmental structures, respectively—to explain how the spread of fake news is exacerbated or attenuated. Recognizing that citizens’ risks and uncertainties fuel the spread of fake news (Matook et al., 2022), we operationalize these factors within the national context. On the supply side, we identify local mobile content and services as a critical factor that provides opportunities for producing and perpetuating fake news (Gupta et al., 2022). On the demand side, we highlight citizens’ subjective well-being and happiness as a proximal factor that can reduce victimization risks and inhibit the spread of fake news. Providing a supportive psychosocial environment characterized by resilience and trust can help mitigate citizens’ susceptibility to misinformation (Matook et al., 2022). These context-specific supply and demand structures serve as critical situational factors influencing the spread of fake news. To counteract negative impacts, governments must address citizens’ risks and uncertainties by providing validated information through trusted local mobile channels. Additionally, improving the overall happiness levels within a country can further minimize victimization risks.

Second, we introduce and theorize the influence of situational preventive structures, specifically the quality of a nation’s legal system. High-quality legal systems have been shown to deter cyberdeviant behaviours by increasing the perceived risks and consequences of deviance, thereby reducing fake news propensity within nations. In our framework, legal guardianship—manifested as a high-quality legal system—serves a moderating role by mitigating the influence of local mobile content and services on fake news propensity. Our findings provide empirical support for our proposed holistic theoretical framework, which integrates demand, supply, and preventive structures to explain fake news propensity at the macro level by leveraging insights from information

systems (IS) and criminology literature. This research advances the theoretical understanding of fake news propensity.

Third, our work introduces a novel theoretical perspective in the respect that, rather than focusing solely on victimization levels, it examines the supply and demand factors that underpin cyberdeviant behaviour and thus contribute to fake news propensity. This approach distinguishes our research from prior studies of cyber deviance within IS (Cheng et al., 2020) and beyond (Williams, 2016). Moreover, our theorizing and empirical testing of the macro-level influence of these factors on deviance-related outcomes means that our study addresses recent calls from outside IS for further research (Olson et al., 2021). By exploring the relationship between societal happiness levels and cyberdeviance in the context of fake news production and dissemination, we provide a practical and quantifiable framework to operationalize citizens’ cognitive mindsets at the societal level. Furthermore, our theorization demonstrates how these cognitive mindsets interact with technological structures to mitigate opportunities for cyberdeviance. This operationalization offers researchers a foundation to investigate other forms of cyberdeviance for IS scholars, at the macro level, such as cybersecurity threats, cyberbullying, and conspiracy-driven radicalization (Abdalla Mikhael & Baskerville, 2023; Naidoo, 2020).

Fourth, our perspective on cyberdeviant behaviour, analyzed through the spread of fake news, provides insights through a holistic situational crime prevention lens. Specifically, we examine how supply-side and demand-side factors jointly influence the spread of fake news through both direct and moderated pathways, moving beyond the conventional focus on victimization risk reduction. And we highlight the critical role of legal guardianship in moderating the influence of supply-side factors, such as the spread of fake news via local mobile content and services. In particular, a stricter rule of law can serve as a significant barrier to cyberdeviant actors, deterring them from exploiting technological resources to produce and disseminate fake news. In so doing, this research contributes valuable policy-level insights for mitigating the global risks posed by the spread of fake news.

Fifth, our research provides valuable insights into the supply and demand factors that drive the production and consumption of fake news within nations. The availability of situational opportunity structures, such as localized mobile content and services supported by mobile technology infrastructure, is positively associated with the spread of fake news because it enables easy access to potential victims. By contrast, situational psychosocial environmental structures, particularly citizens’ cognitive framing and happiness levels, are negatively associated with the spread of fake news (Lowry et al., 2016). Additionally, the quality of

a nation's legal system significantly influences fake news propagation. Consistent with prior non-IS research emphasizing the importance of high-quality legal structures to curb misinformation (Alemanno, 2018), our findings validate the effectiveness of the “rule of law” perspective. Specifically, situational preventive structures such as legal guardianship play a critical role in deterring potential cyberdeviants from spreading misinformation. The various boundary conditions uncovered in this study can act as a starting point for reconceptualization of mobile regulation and/or for creating new fake news regulation, as it can influence the collective prospective sensemaking processes that are said to occur between diverse stakeholders while regulating emerging technology (Seidel et al., 2025), by extension, a sociotechnical phenomenon such as cyberdeviance or fake news.

9.2. Limitations

While this sequential mixed-methods study offers theoretical and practical contributions to addressing the complex phenomenon of fake news, several limitations must be acknowledged.

First, the study was conducted during a global crisis, which may have significantly influenced public cognition and media consumption patterns. The crisis offers a natural experiment setting where all nations across the world were faced with a similar pandemic situation. Hence, the object of fake news was similar, providing conditions for a meaningful cross-country comparison. The key structural conditions contributing to fake news deviance, as per our model, remain consistent across contexts. These include the availability of contextualized content and supporting technological infrastructure, the psychosocial environment in which deviance flourishes/emerges, and the presence or absence of effective control structures such as quality legal systems. These factors are not unique to crisis periods and are also present in non-crisis contexts. Although such effects may become more salient or accelerated during crises, making the findings transferable, provided that appropriate contextual controls are included in the model to account for variations across specific settings. As such, the applicability of the proposed theoretical model under normal, non-crisis conditions requires further empirical validation.

Second, national-level fake news propensity was proxied using reports from centralized fact-checking organizations. However, such data may be shaped by disparities in institutional capacity, resource availability, and reporting emphasis. Only well-resourced fact-checkers may have been able to systematically identify and report fake news, potentially skewing the observed levels across countries.

Third, while our dataset spans 65 nations, the cross-national spread and transboundary influence of fake news, which is an increasingly global concern, was not

directly examined in this study. This limitation underscores the need for future research that captures the interconnected nature of misinformation and fake news flows across borders.

Fourth, the qualitative phase, which involved interviews with eight senior domain/policy experts, provided rich insights into contextual boundary conditions. Nevertheless, the perspectives of sub-national actors or broader policy communities remain underexplored, and incorporating more diverse policy voices could further enhance the representativeness and transferability of the findings.

Finally, this study draws on validated secondary cross-national indicators that capture aggregated structural and institutional conditions and may not fully reflect short-term societal or policy changes within countries. The analysis focuses on macro-level factors and does not preclude the influence of individual- or group-level determinants of fake news vulnerability. Moreover, given the cross-sectional design, the findings should not be interpreted as causal. From a policy standpoint, restricted digital access to local content and services or stronger legal institutions alone may be insufficient to curb fake news, as outcomes can also be shaped by mobile media platform governance and public trust.

9.3. Concluding remarks

Framing fake news proliferation as a manifestation of cyberdeviance, our study conceptualized the situational country-level factors that augment or inhibit fake news deviance within nations. We found that local mobile content and services create technological opportunities for producing and disseminating fake news, while citizens' happiness levels reduce the demand for and spread of misinformation by fostering resilience against it. Additionally, the quality of legal guardianship within a nation mitigates the influence of technological opportunities on fake news propensity. Our study contributes to the literature by applying a crime-theoretic perspective to identify and regulate the growing challenges posed by fake news. Beyond developing and empirically validating a theoretically driven model, we identified important boundary conditions that provide valuable policy insights for governments.

We urge governments to consider the results and insights from this study to design holistic prevention strategies that build long-term public sector resilience against fake news propagation. We also encourage future researchers to incorporate theories from criminology, well-being, and communication policy into fake news research. As we have shown, doing so can yield theoretically grounded and practically

meaningful insights, helping to address the growing menace of fake news.

Notes

1. The term cyberdeviance is used as opposed to deviance or crime because fake news in our research context reflects a digitally mediated norm violation shaped by national socio-technical conditions, rather than a legal or purely communicative phenomenon.
2. All boldfacing here and elsewhere has been done by the authors to emphasize the most relevant portions of the interview quotes.
3. We thank one of the anonymous reviewers for suggesting this point.
4. However, such a design often depends on the strength of civil society and media ecosystems, which some scholars fear can be eroded by the economics of social media and prevalence of various types of AI-generated content, including deepfakes that are not easily distinguishable by an expert or a layperson (Shirish & Komal, 2024).
5. We thank one of the anonymous reviewers for suggesting this addition.

Author contributions

CRediT: **Anuragini Shirish:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Validation, Writing – original draft, Writing – review & editing; **Shirish C. Srivastava:** Conceptualization, Funding acquisition, Supervision, Validation, Writing – review & editing; **Shalini Chandra:** Data curation, Formal analysis, Validation, Writing – review & editing.

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Appendices

Appendix A. List of variables

Fake News Propensity refers to the volume of COVID-19 related fact-checks on misinformation originating from each country in the global database of International Fact-Checking Network (IFCN), Poynter Institute as of June 9 2020. The volume of fake news is measured using the numeric "count data" available from the Poynter website. Source link: <https://www.poynter.org/ifcn-covid-19-misinformation/>

Independent Variables

Mobile Content and Service refers to the availability of secure online content and services accessible and relevant to the local population in a country. It is indicated by the Mobile Content and Services Sub Index for each country from the Global System for Mobile Communications Association (GSMA Intelligence) Foundation Intelligence Report 2020. The index assesses the performance of countries on the key enablers such as local relevance (top level domain name per capita, social media penetration, mobile application development), availability (Number of apps in national language and access to top ranked apps) and online security (global cyber security levels). The value is scored on a numeric scale in the range "0" to "100", where a higher score represents stronger performance in delivering mobile content and services. Source link: https://www.mobileconnectivityindex.com/?source=post_page-----#year=2019&dataSet=indexScore

Subjective Well Being/Happiness Index is used to measure the collective happiness and well-being of a population using a bottom-up approach based on subjective perceptions of their life evaluation. It is based on a Cantril ladder survey. Nationally representative samples of respondents are asked to think of a ladder, with the best possible life for them being a 10, and the worst possible life being a 0. They are then asked to rate their own current lives on that 0 to 10 scale. The data is drawn from the Gallup World Poll. The level of happiness is measured on a scale ranging from "1" to "10" with 1 indicating lower degree of subjective wellbeing and "10" indicating "the highest degree of subjective well-being". Source link: It is published by United Nations Sustainable Development Solutions Network in their Happiness Report 2020. Source link: <https://worldhappiness.report/ed/2020/>

Rule of Law refers to the extent to which there are legal guarantees of rights and on the ground fulfillment provisions. Thus, both law and actual practices are factored into the scoring, the emphasis is on implementation of laws. It takes into account if the nation has an independent judiciary; if there is protection from the illegitimate use of physical force and freedom from war and insurgencies. The values have been taken from freedom in the World Index Report 2020. The level of rule of law is measured on a scale ranging from "1" to "15" with 1 indicating lower degree of rule of law and "15" indicating "the highest degree of rule of law". Source link: <https://freedomhouse.org/report/freedom-world/2020/leaderless-struggle-democracyorg/index/about>.

Control Variables

Gross Domestic Product (GDP) represents the standard of living in a country and is related to its productivity. The values of GDP in US Dollar (in billions) are taken from Global Health Security Index 2019 Report, which provides these figures compiled from the Economic Intelligence Unit, World Bank, and CIA World Factbook 2019. Source link: <https://www.ghsindex.org/>

Unemployment Rate represents the number of unemployed persons measured as a percentage of the total labor force. The data has been taken from the Heritage Foundation Report 2020, which has compiled the most recent unemployment rate data from International Labour Organization and World Employment Social Outlook: Trends 2019. Source link: <https://www.heritage.org/index/about>

Quality of Healthcare is indicated using the quality of the healthcare index from the Global Health Security Index 2019 Report. It is a composite of three indicators (1) life expectancy measured in years, (2) healthcare access, and (3) quality index frontier score. The construct is measured on a numeric scale ranging from 0 to 100, with "0" indicating "lowest quality of healthcare" and "100" indicating "highest quality of healthcare". Source link: <https://www.ghsindex.org/>

Literacy Rate is indicated by the literacy levels index from the United Nations Development Programme and the UNESCO Institute for Statistics (UIS) Reports 2019. It is measured as the percentage of literate adults (above 15 years) in a country. Source link: <http://hdr.undp.org/en/data>

Economic Freedom refers to the fundamental right of every human-being to control his or her own labor and property in a country. The construct "Economic Freedom" is modelled as a single indicator index comprising twelve quantitative and qualitative factors grouped into four broad categories related to the level of economic freedom in a country, namely -(1) rule of law, which includes the property rights, government integrity, and judicial effectiveness (2) government size, which includes the government spending, tax burden, and fiscal health (3) regulatory efficiency, which includes the business freedom, labor freedom and monetary freedom (4) open markets, which includes the trade freedom, investment freedom, and financial freedom. The values have been taken from the Heritage Foundation Report 2020 where countries based on a composite score are grouped on an economic freedom scale ranging from "1" to "5", with "1" indicating "repressed" and "5" indicating "free". Source link: <https://www.heritage.org/index/about>

Political Freedom refers to the freedom related to citizen's political choice and participation in a country. The construct "Political Freedom" is modelled as a composite country-level index comprising three indicators related to the level of freedom in (1) electoral process (2) political pluralism, and (3) participation in the functioning of the government. The values have been taken from freedom in the World Index Report 2020. The level of political freedom is scored on a numeric scale ranging from "1" to "7" and reverse coded, with "1" indicating "the lowest degree of political freedom" and "7" indicating "the highest degree of political freedom". Source link: <https://freedomhouse.org/report/freedom-world/2020/leaderless-struggle-democracy>

Media Freedom refers to the degree of freedom that journalists, news organizations, and netizens have in each country, and the efforts made by authorities to ensure this freedom. The construct "Media Freedom" is measured using Press Freedom Index, the values for which are taken from Reporters Without Borders Report 2020. The index comprises media pluralism, independence, legislative quality related to media and safety of journalists. The level of media freedom is based on scale ranging from "1" to "5" and reverse coded, with "1" indicating "no media freedom" and "5" indicating "complete media freedom". Source link: <https://rsf.org/en/ranking>

Mobile Infrastructure refers to the availability of high-performance mobile internet network coverage. It is indicated by the Mobile Infrastructure Sub Index for each country from the Global System for Mobile Communications Association (GSMA Intelligence) Foundation Intelligence Report 2020. The index assesses the performance of countries on the key infrastructure enablers such as network coverage (2 g to 5 g network coverage), network performance (mobile upload, download speeds, latencies), other enabling infrastructure (such as electricity, number of internet exchange points, international bandwidth per user) and spectrum levels. The value is scored on a numeric scale in the range "0" to "100", where a higher score represents stronger performance and presence of mobile infrastructure. Source link: https://www.mobileconnectivityindex.com/?source=post_page-----#year=2019&dataSet=indexScore

Mobile Affordability refers to the availability of mobile services and devices at price points that reflect the level of income across a national population. It is indicated by the Mobile Affordability Sub Index for each country from the Global System for Mobile Communications Association (GSMA Intelligence) Foundation Intelligence Report 2020. The index assesses the performance of countries on the key enablers such as mobile content tariff, mobile handset price, taxation on mobile sales and spectrum taxes, and inequality in income levels. The value is scored on a numeric scale in the range "0" to "100", where a higher score represents stronger performance in delivering mobile content and services. Source link: https://www.mobileconnectivityindex.com/?source=post_page-----#year=2019&dataSet=indexScore

Appendix B. Descriptive statistics

	Mean	Std. Deviation	N
VFN	92.42	207.042	65
MIF	64.877	14.318	65
MAF	59.833	14.388	65
GDP	1144.950	3077.808	65
LIT	85.885	19.247	65
UNE	6.075	4.676	65
PFD	4.7385	2.094	65
EFD	2.877	.976	65
MFD	2.892	1.226	65
QHC	60.275	22.964	65
HAP	5.703	1.106	65
MCS	128.650	38.842	65
ROL	8.370	4.602	65

Notes: VFN: Volume of Fake News; MIF: Mobile Infrastructure; MAF: Mobile Affordability; GDP: Overall Gross Domestic Product; LIT: Literacy; UNE: Unemployment; PFD: Political Freedom; EFD: Economic Freedom; MFD: Media Freedom; QHC: Access to Quality Health Care; MCS: Mobile Content and Services; ROL: Rule of Law; HAP: Happiness.

Appendix C. Correlations

	VFN	MIF	MAF	GDP	LIT	UNE	PFD	EFD	MFD	QHC	MCS	FROL	HAP
VFN	1	.037	.084	.373**	.057	-.079	.153	-.210	-.004	.063	.122	.120	-.044
MIF	.037	1	.742**	.070	.080	.061	.122	.388**	.104	.189	.871**	.632**	.753**
MAF	.084	.742**	1	.065	.033	-.093	.130	.406**	.150	.173	.718**	.587**	.748**
GDP	.373**	.070	.065	1	.190	-.054	.082	-.056	-.050	.255*	.173	.085	.164
LIT	.057	.080	.033	.190	1	-.012	.118	.081	.083	.775**	.025	.135	.085
UNE	-.079	.061	-.093	-.054	-.012	1	.266*	-.050	.200	.066	.027	.158	.011
PFD	.153	.122	.130	.082	.118	.266*	1	.397**	.810**	.061	.103	.223	.164
EFD	-.210	.388	.406**	-.056	.081	-.050	.397**	1	.445**	.142	.246*	.379**	.290*
MFD	-.004	.104	.150	-.050	.083	.200	.810**	.445**	1	.064	.069	.276*	.192
QHC	.063	.189	.173	.255*	.775**	.066	.061	.142	.064	1	.130	.218	.153
MCS	.122	.871**	.718**	.173	.025	.027	.103	.246**	.069	.130	1	.622*	.782**
ROL	.120	.632**	.587*	.085	.135	.158	.223	.379*	.276*	.218	.622**	1	.645**
HAP	-.044	.753**	.748**	.164	.085	.011	.164	.290**	.192	.153	.782**	.645**	1

Notes: VFN: Volume of Fake News; MIF: Mobile Infrastructure; MAF: Mobile Affordability; GDP: Overall Gross Domestic Product; LIT: Literacy; UNE: Unemployment; PFD: Political Freedom; EFD: Economic Freedom; MFD: Media Freedom; QHC: Access to Quality Health Care; MCS: Mobile Content and Services; ROL: Rule of Law; HAP: Happiness.

**Correlation is significant at the 0.01 level (2-tailed). The main variables are centered.

*Correlation is significant at the 0.05 level (2-tailed).